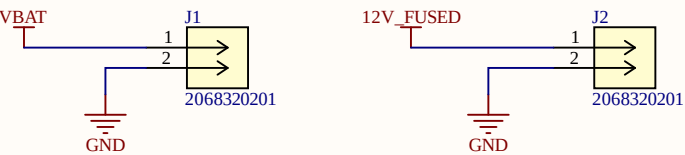
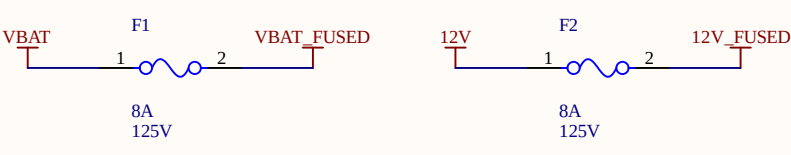


CONNECTORS

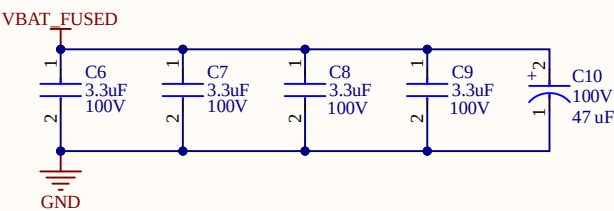


FUSE HOLDERS

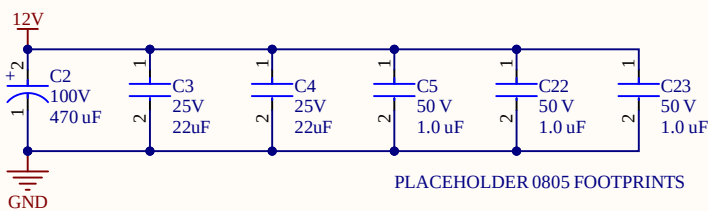


FUSES MUST BE RATED FOR 12A ON OUTPUT, 5A ON INPUT
INPUT CURRENT IS ~2.5A AT MAXIMUM LOAD CURRENT OF 9A, ASSUMING 90% EFFICIENCY

INPUT FILTERING



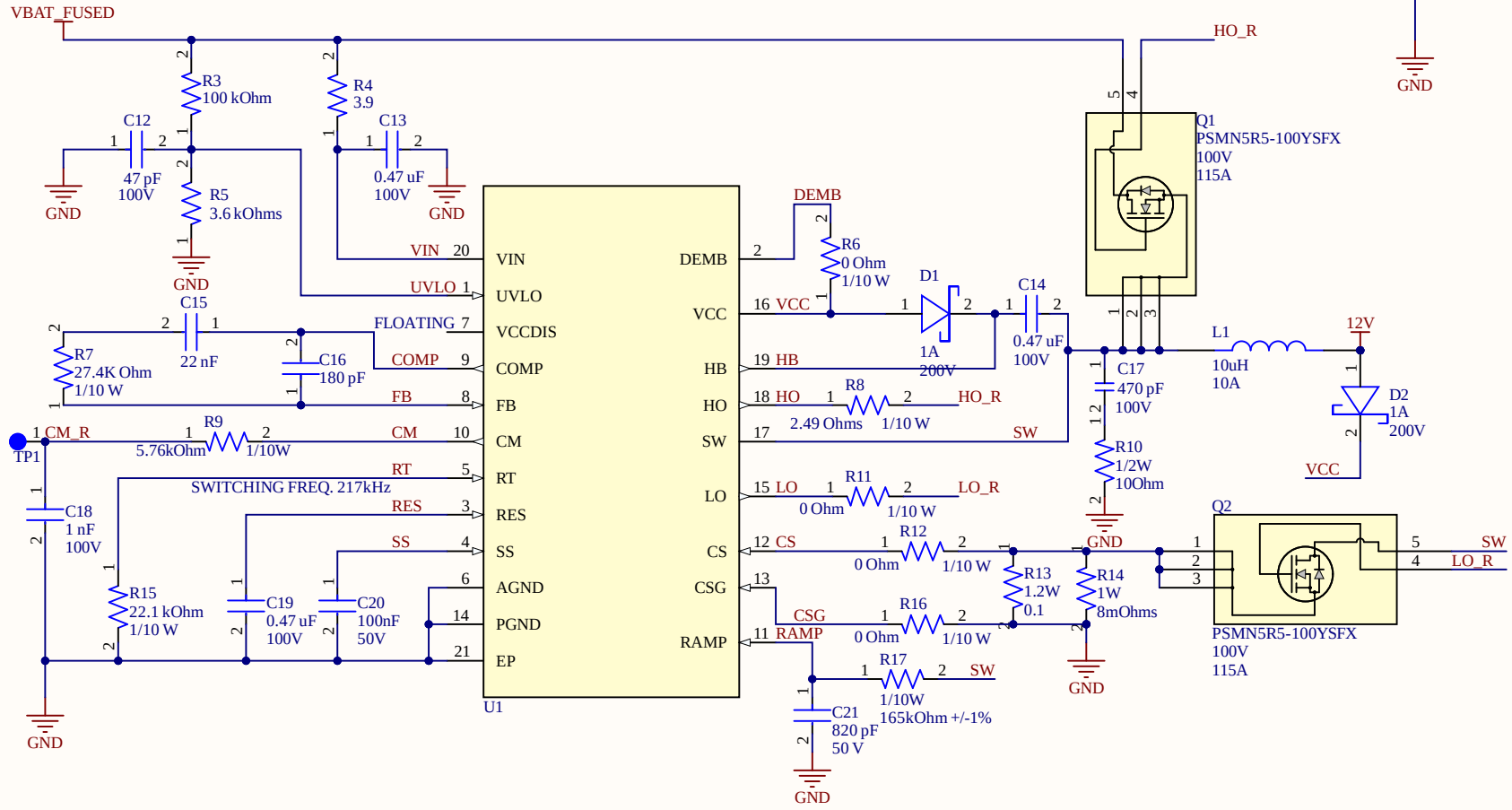
OUTPUT FILTERING



PLACEHOLDER 0805 FOOTPRINTS

DEMB- WHEN LOW, EFFICIENCY BETTER, BUT WORSE TRANSIENT RESPONSE (SWITCHING FREQ. VARIES)
VCC DRIVES FET GATES, DIODE SO NO CURRENT FLOWS INTO VOUT
HO/LO HAS DAMPING RESISTORS TO CREATE RLC DUE TO HIGH SWITCHING
CS (CURRENT SENSE) MEASURES WHEN LO SIDE IS ON, TRIPS AT 16.4A (7.3mOhms Req)
RAMP PIN USED TO EMULATE L CURRENT USING VOLTAGE ON EITHER SIDE OF L

UVLO SIZED FOR 36V TO ACCOMODATE BATTERY VOLTAGE CHANGES, 2V HYSTERESIS ON BOOT
VIN HAS LPF WITH R4 AND C13, CUTOFF ~87kHz TO ELIMINATE SWITCHING FREQ
VCCDIS HAS INTERNAL PULLDOWN TO ENABLE INTERNAL LDO/EXTERNAL 12V ONCE STABLE
COMP-FB CIRCUIT IS USED TO ENSURE BUCK RESPONDS TO DC CHANGES IN OUTPUT, DOESN'T OVERCORRECT TO TRANSIENTS
CM (CURRENT MONITOR) HAS LPF TO CUTOFF SWITCHING NOISE
RES & SS DETERMINE HICCUP PERIOD & RISE TIME



Name	Description	Designator	Footprint	Library	Quantity
ES9477M100AM02A	CAP ALUM 470UF 20V 100V NACA	C2	CAP-470U-0001	CAP-470U-0001	1
CRMS20R1E220K15	CAP CER 22UF 25V 10% 1210	C3, C4	CAP-1210	CAP-22U-0001	2
CC0806XK7988B105	CAP CER 10UF 50V X7R 0603	C3, C22, C23	CAP-0603	CAP-1U-0001	3
CC3216V752A33K160A	CAP CER 3.3UF 100V X7S 1206	C8, C7, C8, C9	CAP-1206	CAP-3.3U-0001	4
LUJ24470MNL1G5	47 uF 100 V Aluminum Electrolytic Capacitor Radial Can -84D 2000 Hrs @ 105°C	C10	PCAP_10x10-ELECT_NCA	CAP-47U-0005	1
UNMK107AB7105KA-1	1 uF ±10% 50V Ceramic Capacitor X7R 0603 (1608 Metric)	C11	CAP-0603	CAP-1U-0002	1
744778B111	47 pF ±5% 100V Ceramic Capacitor C0G, NPO 0603 (1608 Metric)	C12	CAP-0603	CAP-47P-0001	1
KAM21K72A474K5	0.47 uF ±10% 100V Ceramic Capacitor X7R 0805 (2012 Metric)	C13, C14, C19	CAP-0805	CAP-0.47U-0001	3
C0603C22K39AC786 7	0.022 uF ±10% 25V Ceramic Capacitor X7R 0603 (1608 Metric)	C15	CAP-0603	CAP-22N-0001	1
CC0603RNPO88N131	180 pF ±5% 50V Ceramic Capacitor C0G, NPO 0603 (1608 Metric)	C16	CAP-0603	CAP-180P-0001	1
0603B471K101CT	470 pF ±10% 100V Ceramic Capacitor X7R 0603 (1608 Metric)	C17	CAP-0603	CAP-470P-0001	1
C0603Y10DK39AC786 7	1000 pF ±10% 100V Ceramic Capacitor X7R 0603 (1608 Metric)	C18	CAP-0603	CAP-1N-0001	1
CC0603K0K7988B104	0.1 uF ±10% 50V Ceramic Capacitor X7R 0603 (1608 Metric)	C20	CAP-0603	CAP-100N-0001	1
C0603C21K39ACTU	0.021 uF ±10% 10V Ceramic Capacitor X7R 0603 (1608 Metric)	C21	CAP-0603	CAP-823P-0001	1
BSL1200	DIODE SCHOTTKY 100V 1A SMA	D1, D2	SMA_DIO	DI0-SCHOTTKY-0004	2
01550900M	FUSE BLOW CARTRIDGE 125V 10A 5A5	FL F2	FUSE-0001	FUSE-0001	2
2068320201	Micro-Fin Vertical Header, Through Hole, 0.80mm Pitch, Dual Row, 2 Circuits, Tin [Sn] Plating, Glow Wire Capable, Black, Tray	U1, U2	CONN_2068320000-02_QC_MSL	2068320201	2
SRP1265A-100M	10 uH Shielded Drum Core, Wirewound Inductor 10 A 1% SMD Core Max Nonstandard	L1	IND_SRP1265A-100M	IND4ENC	1
PM5R5R-10019FX	15 Ohms ±1% 0.1W (1/10W) (F1) Surface Mount LPM5R5, Power SMD	Q1, Q2	LPM5_10T669_NEX	PM5R5-115A-0001	2
RC0603FR-0714K1	14 kOhms ±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R1	RES-0603	RES-14K-0001	1
RT0603FRE131K1	1 kOhms ±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Thick Film	R2	RES-0603	RES-1K-0002	1
RC0603FR-07100K1	100 kOhms ±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R3	RES-0603	RES-100K-0003	1
RC0603FR-0739R1	0.3 Ohms ±1% 0.12W, 1/10W Chip Resistor 0603 (2012 Metric) Moisture Resistant Thick Film	R4	RES-0805	RES-3.9-0001	1
RC0603FR-0736K1	0.6 kOhms ±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R5	RES-0603	RES-3.6K-0001	1
RMCF0603Z10R00	0 Ohms Jumper 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Automotive AEC-Q200 Thick Film	R6, R11, R12, R16	RES-0603	RES-0-0001	4
RC0603FR-0727K4L	27.4 kOhms ±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R7	RES-0603	RES-27.4K-0001	1
RC0603FR-0724K91	2.49 kOhms ±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R8	RES-0603	RES-2.49-0001	1
RG03FTC5761	100mW Thin Film Resistor 75V ±25ppm/°C ±1% -55°C~+155°C S, 7642 0603 Chip Resistor Surface Mount R06G	R9	RES-0603	RES-5.76K-0001	1
S073P1J7TD1000	10 Ohms ±5% 0.5W, 1/2W Chip Resistor 0603 (1608 Metric) Automotive AEC-Q200, Pulse Withstanding Thick Film	R10	RES-0603	RES-10-0001	1
PS0508FRE070K1L	100 mOhms ±1% 1.2W Chip Resistor M40K 0805 (2012 Metric), 0808 Automotive AEC-Q200, Current Sense Metal Element	R13	RES-0805	RES-0.1-0001	1
PL1206PM470R00B2	0 mOhms ±1% 12W Chip Resistor 1206 (3224 Metric) Current Sense Metal Foil	R14	RES-1206	RES-0mOhms-0001	1
RC0603FR-0722J1L	22.1 kOhms ±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R15	RES-0603	RES-22.1K-0001	1
RC0603FR-07169K1	169k Ohms ±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R17	RES-0603	RES-169K-0001	1
S003	Chromic PC Test Point, Miniature Phosphor Bronze Silver Plating Through Hole Mounting Type	TP1, TP2, TP3, TP4, TP5, TP6, TP7, TP8, TP9, TP10, TP11, TP12	TEST-POINT-0001	TEST-POINT-0001	12
LM5117PM4U/NOPB	Black Regulator Positive Output Step Down DC-DC Controller IC 20-HTSOP	U1	IC-20-HTSOP	IC-BUCK-0011	1